

IBM Mainframes

History, Architecture, Business Uses, and Operating Systems





WHAT IS A MAINFRAME, ANYWAY?

The word *mainframe* generally refers to any large computer. There's an implied sense of *high capacity* and *centralized computing*.

Historically, the term comes from the fact the components of early computer systems were so bulky that they had to be installed in custom-designed racks called *frames*.

The frame that housed the central processing unit (CPU) was called the *main* frame.

Other frames housed peripheral devices and memory units. Each of these things was furniture-sized. A tape drive of the era was larger than a household refrigerator.

Today's mainframes are based on the same technologies as other computer systems and can fit into standard 19-inch server racks (between 1 and 4 racks per system).

They can still run the same software as mainframes from the 1960s, along with more contemporary software.



IBM 703, 1953
1 processor, 9 KB memory

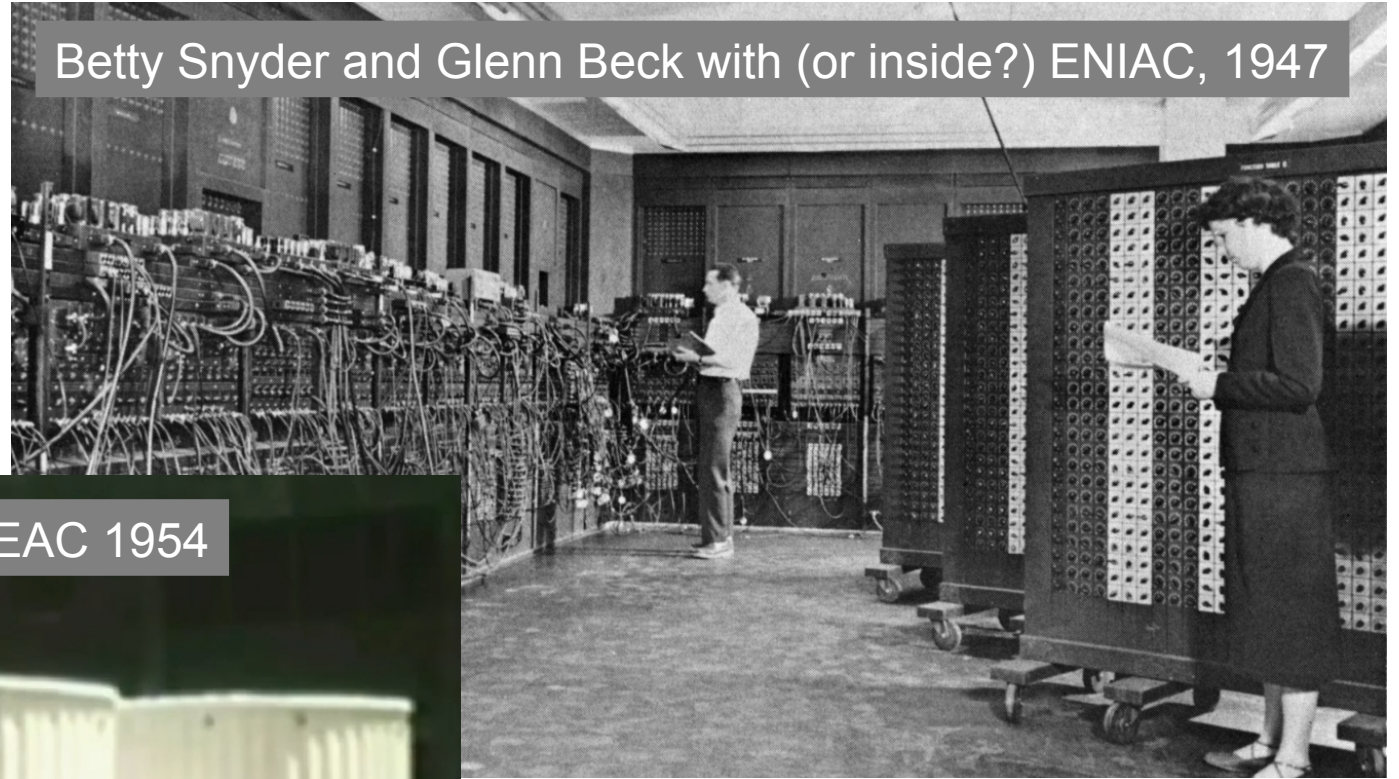


IBM z16, 2023
192 processors, 42 TB memory

HISTORY

Computer systems were first used by the military in the late 1940s to calculate trajectories for ballistic missiles and artillery. Each computer was a custom build.

Betty Snyder and Glenn Beck with (or inside?) ENIAC, 1947



First portable computer, DYSEAC 1954



Their usefulness was immediately apparent to business leaders. Demand for mass-produced computers grew quickly. The 1950s and 1960s saw intense competition and rapid technological development in computer systems.

HISTORY

We can think of the history of mainframes as consisting of three stages. The names of these stages are my own, coined for the purpose of this presentation. The date ranges are approximate and overlapping.

Initial stage – *rapid innovation, intense competition, market shake-out*

1950s – 1970s

Many companies enter the market and several prosper, leap-frogging each other with innovations that drive computer technology forward rapidly.

Survival stage – *IBM seeks a way forward for mainframes in the face of newer technologies*

1980s – 1990s

As people find cost-effective alternatives to low-end and mid-tier mainframe systems, and *Information Week* predicts the last mainframe will be unplugged in 1996, IBM looks for ways to keep the systems relevant.

Modernization stage – *Significant development of mainframe technology and software*

2000s - present

A thorough and very impressive refresh of the mainframe from the hardware up made the system the platform of choice for operations calling for extreme scale, high security, high performance, high reliability, and backward compatibility.

HISTORY: INITIAL STAGE 1950-1980

Although many companies made a go of building and selling computer systems, IBM's main competition came from five companies collectively called *The Bunch*, based on the first letters of their names:



The competition was cut-throat and makes for an interesting tale in its own right. But that's out of scope for this training.

Suffice it to say that by 1980 there was only one significant player in the mainframe space: IBM.

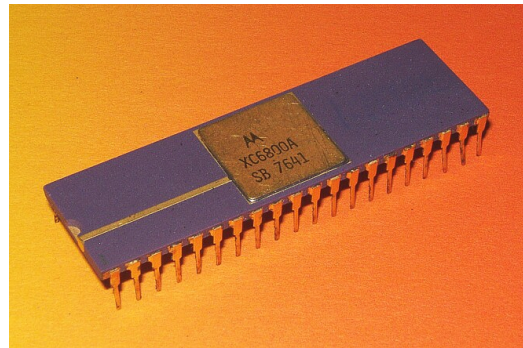
Compatible machines were offered by Amdahl, National Semiconductor, Fujitsu, and Hitachi, but they all ran IBM software and were based on IBM's design.

HISTORY: SURVIVAL STAGE 197X-2000

The push for innovation from the end of World War II until the invention of the personal computer around 1980 resulted in a level of technological change that altered the landscape of the computer industry. Such things as semiconductors and symmetric multiprocessing systems, as well as innovation in operating systems that enabled virtual memory and multithreading, opened the door for new competitors to IBM.



DEC VAX 11/780, 1977



Motorola 6800, 1974



Commodore Pet, 1983

Companies like Digital Equipment Corporation (DEC) and Wang offered *minicomputers* – systems capable enough to handle the workload of a small to mid-sized IBM mainframe at far less cost. The advent of even smaller hardware brought *microprocessors* to the market, enabling a lifestyle-changing innovation: The personal computer.

HISTORY: SURVIVAL STAGE 197X-2000

Possibly the biggest threat to IBM's dominance of the business computing market was a shift in mindset away from centralized computing and toward networked computing.

Sun Microsystems appeared in 1982 with the slogan, "The network is the system." Tim Berners-Lee invented the World Wide Web between 1989 and 1993.

People predicted the end of the mainframe era, and many projects were undertaken to migrate critical business applications from the mainframe platform to other systems.

Fortunately for IBM, this proved far more difficult and costly than anyone expected. They had some breathing space, but no time to waste.



Sun server racks, Seneca College, NY

HISTORY: SURVIVAL STAGE 197X-2000



IBM RS/6000s, 1990s

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IBM faced one more tough competitor: Itself.

In the 1990s, IBM had multiple products that targeted the same market:

- IBM 4300 series mainframes
- IBM AS/400
- IBM System/36 and 38
- IBM RISC System 6000

They were also trying to compete in the personal computer space, which was not their strong suit – their customer base consisted of large corporations and government agencies.

HISTORY: MODERNIZATION STAGE 199X-PRESENT

Under new leadership, IBM dropped some product lines and focused on their mainframe and midrange business computer systems:

- IBM System/390, which would go through several new names before settling on "IBM z".
- IBM Power Systems, the direct descendant of the AS/400

They sold off the consumer product line to Lenovo.

The company concentrated its efforts on the market segment where it was still dominant: The very largest mainframe installations, supporting the heaviest workloads and most stringent security requirements.

They added support for POSIX, Java, C++, TCP/IP, and Linux. They beefed up security, performance, and reliability. They added support for DevOps, cloud, Docker, and Kubernetes.

And they maintained their commitment to backward compatibility and standards.



IBM zSeries, circa 2000

MAINFRAMES TODAY

The result of all these changes is the system you're learning to work with today. It's not so much a single system as a network in a box that supports just about every computing technology that's come along since 1964, all at the same time.

The backward compatibility aspect is what makes the system a bit intimidating to get started with. Not only does the system support many different technologies, but it supports several versions of each technology.

It also enables old and new solutions to be integrated with each other, so that older applications need not be rewritten – a task that has defeated multiple generations of software engineers.

BUSINESS USES

In the heyday of mainframes they were used for every business data processing need. Originally they were batch processing systems. They ran payroll, general ledger, inventory, and a host of other back office applications for enterprises of all kinds and sizes.

But very early in the history of mainframes, certain customers demanded support for interactive processing, too.

- Travel and hospitality – travel and hotel reservations
- Manufacturing – automated factory operations
- Shipping and logistics – realtime management of operations
- Energy – power plant operations

IBM responded to customer needs either by developing custom interactive systems or by adding support for interactive processing on top of their batch-oriented operating systems in the form of long-running jobs that monitored terminals and responded to messages.

BUSINESS USES

For the travel and hospitality industry, IBM developed custom solutions for Delta Airlines and American Airlines. These were later consolidated into the Airline Control Program (ACP), which ultimately evolved into Sabre and the Transaction Processing Facility (TPF).

For other industries, the approach was to run a "batch job" that didn't process sequential files in the way a "normal" batch job did, but instead monitored terminal devices and responded to messages from those devices. To the end users, the application appeared to be interactive.

Mainframes today support core, mission-critical functionality for national banks, large commercial banks, shipping and logistics companies, energy companies, manufacturing operations, reservations, credit authorization, and more.

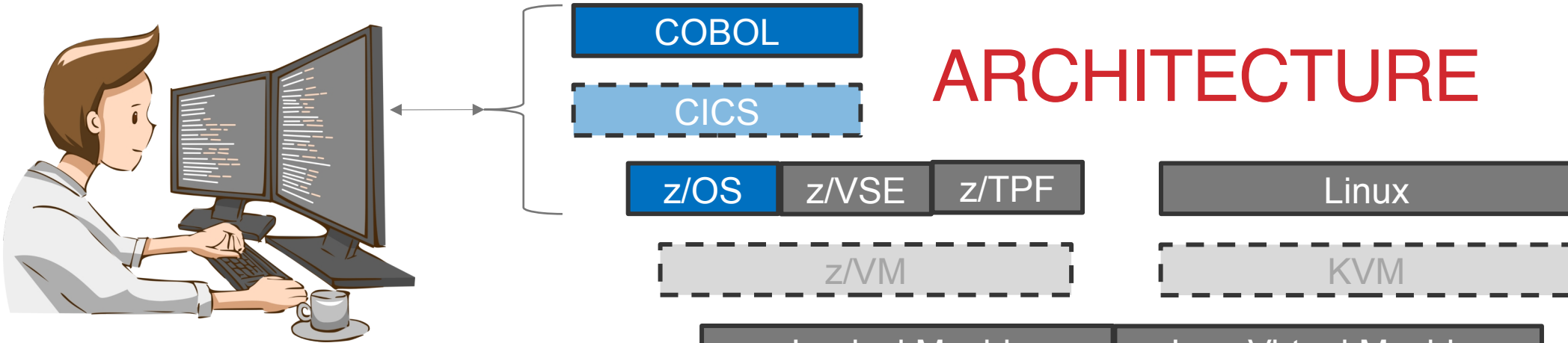
The back-office applications that were once the bread and butter of mainframes have largely been migrated to other platforms. User-facing front-end applications run on other platforms; the mainframe is a back-end system only.

OPERATING SYSTEMS

The IBM z system can run any or all of the following operating systems concurrently:

- **z/OS** – the flagship operating system for the platform; the basis for modernized functionality. **This is the operating system you will be working with.**
- **z/VSE** – an evolved version of the operating system that used to run on smaller mainframes, supported for backward compatibility
- **z/VM** – the original type 1 hypervisor for the mainframe platform
- **z/TPF** – the evolved version of the Transaction Processing Facility (TPF)
- **Linux** – RedHat Enterprise Linux, SUSE, and Ubuntu
- **KVM** – dedicated Linux Kernel Virtual Machine

ARCHITECTURE



We'll introduce selected details about the IBM z architecture in the context of learning how to work with it. We will not go into the subject in depth.

Almost everything on the system is virtualized. Obviously there have to be physical processors, memory, and I/O channels, but OSes and applications never access the hardware directly. All hardware elements are redundant.

Each LPAR is a hypervisor. Each OS is virtualized, even when an LPAR is running only a single OS.

WHAT'S NEXT

- Overview of the z/OS operating system